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Effect of mixed alkaline earth metal oxides (MgO/BaO) on the structure, thermal, mechanics and dielectric properties of borosilicate glass

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Abstract

In this work, SiO_2-B_2O_3-Al_2O_3 glasses doped with different proportions of alkaline earth metal oxides (MgO and BaO) are prepared by using the melt-quenching method. The effects of alkaline earth metal oxide content on the structure, thermal, optical and dielectric properties of glass are investigated through X-ray diffraction (XRD), differential scanning calorimetry, Fourier transform infrared spectroscopy (FTIR), Raman spectroscopy and impedance analyzer. The XRD results confirm the amorphous nature of the glasses. FTIR and Raman spectra reveal that increasing MgO content led to structural relaxation, with a corresponding reduction in glass transition and melting temperatures due to the disruption of [SiO_4] and [BO_4] units.

Additionally, the dielectric constant increases with MgO content, which can be attributed to the enhanced polarizability and charge density of cations. The incorporation of MgO also improves light transmission across the spectrum. These findings elucidate the role of Mg^{2+} and Ba^{2+} in the structural and property evolution of borosilicate glass, providing a theoretical foundation for optimizing material properties and expanding their applications in optics, electronics, and related fields.

Previous

Next

Keywords

Alkali-earth oxide; High-temperature wettability; Shearing force; Dielectric constant

1. Introduction

Borosilicate glass is a kind of important inorganic amorphous material, possessing exceptional attributes including excellent optical transparency, high resistance to elevated temperatures, chemical stability and commendable mechanical strength [[1], [2], [3]]. It is widely used in optical devices, heat-resistant glass, sealing elements, electronic devices and other fields. In addition, borosilicate glass is also widely used in the field of biomedicine [4]. Its excellent biocompatibility makes it an ideal material for biosensors, drug delivery systems and other fields.

Borosilicate glass is a frequently investigated sealing material due to its advantages in large-scale preparation, ease of sealing, and low cost. It offers superior heat resistance compared to polymers and better electrical insulation than alloys [5,6], making it valuable for electronic packaging, aerospace applications, and tempered glass sealing. Key properties such as wettability, coefficient of thermal expansion (CTE), and shear strength are critical for sealing applications. Mismatches in these properties can result in poor adhesion, reduced heat resistance, and lower mechanical strength [7]. In specialized applications, such as high-frequency circuits on glass fiber substrates, borosilicate glass must also exhibit a low dielectric constant and dielectric loss to enhance signal transmission speed, reduce signal delay, and improve signal-to-noise ratio [8]. Meanwhile, the glass properties are influenced by various trace elements. For example, Liu et al. [9] demonstrated that substituting Na_2O with Li_2O in aluminum borosilicate glasses improved network density and chemical stability without compromising the silicate skeleton, although the dielectric constant remains unaffected by the mixed alkali effect. Li et al. [10] explored the effects of doping aluminum borosilicate glasses with mixed rare earth oxides, finding that equal amounts of CeO_2 and La_2O_3 result in a more compact glass network and optimal dielectric properties. The glass transition temperature (Tg) initially increases and then decreases with variations in non-bridging oxygen content. Wei et al. [11] examined the impact of different SiO_2/B_2O_3 ratios on the network and properties of sealing glass for Al_2O_3 ceramic substrates, observing that a SiO_2/B_2O_3 ratio of 1.82 induced boron anomalies, enhanced crystallinity, and promoted $BaAl_2Si_2O_8$ crystal precipitation, resulting in improved bending strength, Vickers hardness, and sealing performance. While these studies have advanced our understanding of how glass components affect sealing performance, there has been less focus on optical properties, wettability, and mechanical strength. So this work will supplement these aspects of glass properties. In practical applications, the visible light transmittance of glass is typically required to exceed 80% [12]. Ideal wettability is achieved with a contact angle of around 30° between the glass and the substrate. Shear strength is also crucial, with general applications demanding 1–5 MPa, while more rigorous sectors, such as automotive or aerospace, require shear strengths of 10 MPa or higher [13]. Understanding these properties is essential for advancing electronic packaging materials.

In the composition of borosilicate glass, SiO_2 and B_2O_3 act as network-forming substance, while Al_2O_3 serves as an intermediate oxide [14,15]. The incorporation of alkaline earth oxides into the glass system has been demonstrated in several studies to affect the glass transition tendency of the SiO_2-B_2O_3-Al_2O_3 ternary system. BaO and MgO are common intermediate oxides. Their addition will affect the chemical and optical characteristics of the glass, significantly impact the overall characteristics. Generally, these two oxides primarily exist as modifiers within the glass matrix. They interact with other oxides to collectively establish the network structure of the glass [16].

Therefore, it is particularly important to examine how changes in oxide content affect the sealing effectiveness of glass. In this paper, the effects of gradual to complete substitution between MgO and BaO on the network and characteristics of borosilicate glass were systematically studied. By comparing and analyzing the effects of two different components, the underlying mechanism of action in the glass preparation process was revealed, thereby offering novel insights and methods for material design and application in relevant fields.

2. Experiment

2.1. Materials

The high-purity chemical reagents used in the experiments, including SiO_2, H_3BO_3, Al_2O_3, K_2O, ZnO, BaO, and MgO, were all purchased from Sinopharm Chemical Reagent Co., with each oxide having a minimum purity of 99.9%. Deionized water was produced using ULPHW.

2.2. Glass preparation and treatment

Glass samples were made via conventional melt-quenching technology. Furthermore, the incorporation of a minor quantity of ZnO is found to enhance the glass-forming capability and reduce viscosity [17]. The label of the glass samples and their compositions were presented in Table 1.

Table 1. The composition of glasses. Some trace oxides are included in the "others" category.

Composition (wt.%)								
Glass	**SiO_2**	**B_2O_3**	**Al_2O_3**	**K_2O**	**ZnO**	**BaO**	**MgO**	**Others**
H40	53.50	21.3	6	3.5	2.5	8	0	5.2
H41	53.50	21.3	6	3.5	2.5	6	2	5.2
H42	53.50	21.3	6	3.5	2.5	4	4	5.2
H43	53.50	21.3	6	3.5	2.5	2	6	5.2
H44	53.50	21.3	6	3.5	2.5	0	8	5.2

The pharmaceutical raw materials were weighed in calculated proportions, mixed, and manually ground for 10–15 min before being placed in a ceramic crucible. Subsequently, the mixtures were kept in the Muffle at 1450 °C for 60 min. During this period, the glass melt was stirred once to improve clarification and homogenization. The homogeneous molten glass liquid was subsequently emptied into a graphite mold to create a glass block, while the excess melt was rapidly quenched in deionized water for quick solidification. After 10 s, the block sample was removed from the mold and heat-treated in a box furnace within the range of 650 °C–750 °C (the temperature range was determined based on the differential scanning calorimetry (DSC) test results of water-quenched glass powder) for 2h. The annealing process for glass is crucial because it enables gradual cooling, which helps to reduce any remaining stress that may have developed during the glass manufacturing process. Subsequently, the glass block was removed and cooled naturally to ambient temperature, resulting in a transparent glass free of stripes and bubbles. The block was cut into a glass sheet with a diameter of 20mm, a thickness of 2mm, and a glass block measuring 5×5×20mm, for subsequent testing (Fig. 1).

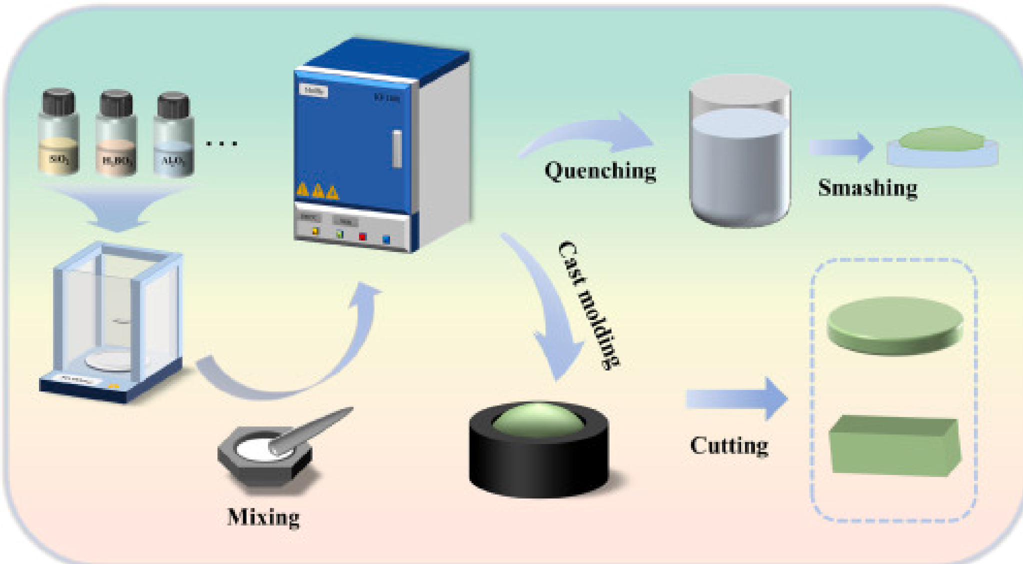

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Fig. 1. The flow chart for sample preparation.

2.3. Analysis methods

The crystal structure was determined by using X-ray diffractometer (XRD, D/Max-3C, Rigaku SmartLab) between diffraction angles from 5° to 80°. The size, morphology and chemical composition of the samples were characterized by field emission scanning electron microscopy (SEM, Quanta 400 FEG instrument, Oxford INCA 35 detector, 25kV). Fourier transform infrared spectroscopy (FTIR, IRTracer-100, Japan) was performed to record the spectra of all the glasses within the 400-4000 cm^{-1} range, and Raman spectra were measured by using a Raman microscope (LabRAM HR JY-Evolution) with 532nm of green laser source.

The Tg was investigated by DSC (Netzsch STA449C, Germany) at 10°C min^{-1} in air. The characteristic temperature points and sintering curves of the glass samples were ascertained by Hot-Stage Microscope (HSM, TA Instruments UB der Waters GmbH, ODP868, S-NO 104014). The wettability between the glass powder and Kovar alloy was determined by contact Angle analyzer (KRUSS, DSA25). The density was calculated by using the Archimedes drainage method.

The transmittance, absorption and reflectivity of the glass sheets in the range of 230–2600nm were recorded by ultraviolet–visible–near infrared spectrophotometer (UV–Vis–NIR, UV-3600 Plus, Japan) after optical polishing of the glass sheet. Thermal expansion curves and CTE of the glass block were decided by using a thermal dilatometer (Netzsch DIL 402CL, Germany). The load-stroke curve of the glass powder bonded Kovar alloy joint was tested by using the universal testing machine (EZ-lx, Shimadzu Corporation, Kyoto, Japan), and the speed of elongation was 2mm per minute. The shear strength τ (MPa) can be calculated by the shear force, by using the formula:

$$\tau = \frac{F}{S} \tag{1}$$

where, F stands for shear force (N) and S stands for effective contact area (mm^2). The glass sheet was sputtered with silver on the surface of the sample by magnetron sputtering under the condition of vacuum 10^{-1} mbar and current 40mA. The dielectric constant and loss were assessed through a wide-ranging dielectric/impedance analyzer (Novocontrol Concept 40, Germany) from 10^0Hz to 10^6Hz. The dielectric constant values, ε_r were calculated using the provided formula:

$$\varepsilon_r = \frac{Cd}{\varepsilon_0 A} \tag{2}$$

where, ε_0 is the dielectric constant of free space (8.854×10^{-12}F/m); The variable d represents the thickness of the glass sheet, while A denotes the area. The formula for determining the dielectric loss value is:

$$tan\ \delta = \frac{G}{\omega c} \tag{3}$$

where, ω=2πf; f is the frequency and G is the conductance.

3. Results and discussion

3.1. Structural characterizations of the glass samples

The SEM images in Fig. 2(a–c) show that the glass powders are composed of irregularly shaped particles with a particle size of about 10–20 μm. The element distribution was analyzed by an energy dispersive X-ray spectrometer, which generated a mapping diagram (d-k). Si, B, Al, O, Mg, Ba, and other elements were uniformly distributed across the glass surface, indicating that the sample glass was successfully prepared. TEM tests were performed on H41. Fig. 3(a and b) depict TEM images of H41 glass sample, which exhibit consistency with the SEM images, all displaying irregular lump-shaped morphology. The HR-TEM image of Fig. 3(c) shows no lattice fringes, and the corresponding SAED pattern in Fig. 3(d) is accompanied by a single diffuse halo, Additionally, the corresponding Fast Fourier Transform (FFT) maps were acquired from the HR-TEM images, as depicted in Fig. S1, with all results indicating an amorphous microstructure.

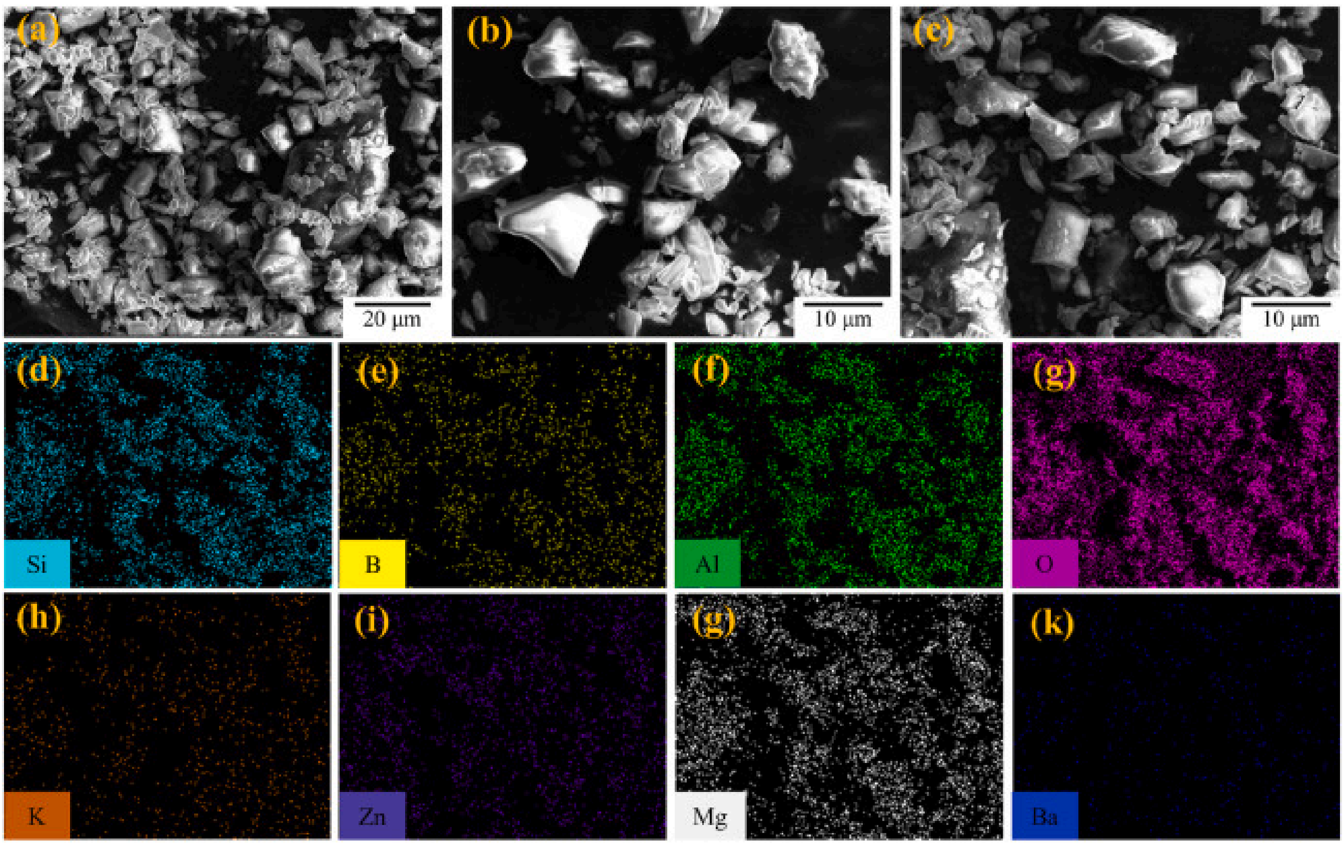

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Fig. 2. (a–c) The SEM of glass samples at different magnifications; (d–k) Mapping analysis results of sample H41 on (c).

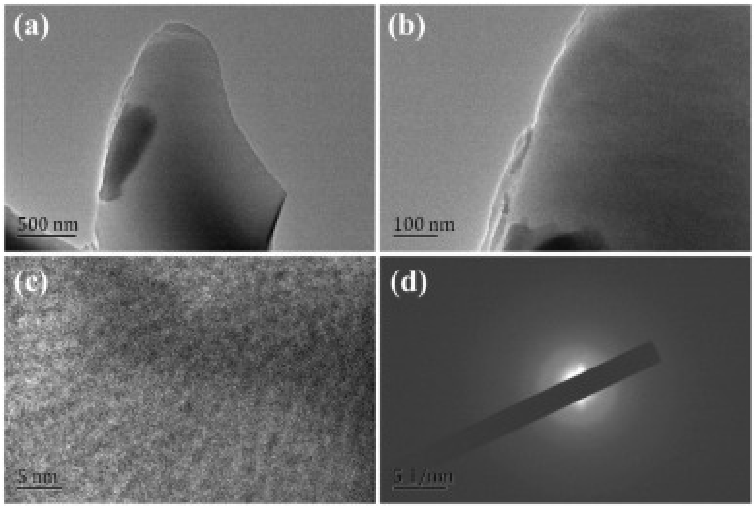

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Fig. 3. (a–b) TEM images of H41 sample; (c) HR-TEM image; (d) SAED pattern.

The XRD patterns of the five glass powders are shown in Fig. 4(a). There are no obvious crystallization peaks at different diffraction angles, which proves the amorphous structure of all samples. There is a broad peak at 21.6°, which is the same as the diffraction peak of SiO_2 [18]. Additionally, the smaller peak around 55° is a typical occurrence [19]. Fig. 4(b) shows the DSC curve of five types of glass powders ranging from 200°C to 1000°C. As the sample temperature increases, the endothermic peak emerges. The curve reveals a decrease in Tg from 460°C to 447°C as MgO concentration increases, and the crystallization temperature (Tc) decreases from 828°C to 796°C. They are all important parameters for characterizing the glassy state. Since both MgO and BaO act as network modifiers, they may have little effect on the Tg of glass with little change in the overall ratio. However, Tc is more sensitive to small changes in composition than Tg, so even small adjustments in oxides may affect the nucleation and growth process of glass. We use Tg and Tc to calculate the value of crystallization resistance of glass, which is also regarded as a common parameter to assess the thermal endurance and stability of glass. The formula is:

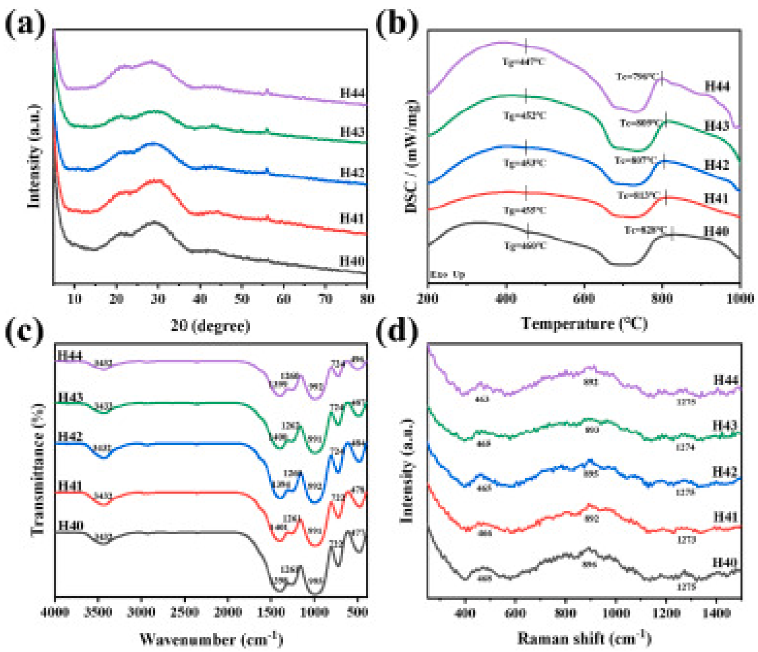

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Fig. 4. (a) XRD patterns of the five samples; (b) DSC curves of the five samples; (c) FTIR spectra of the five samples; (d) Raman spectra of the five samples.

$\Delta T = Tc - Tg$ (4). Table 2 lists the results of Tg, Tc and ΔT. As MgO content rises, ΔT exhibits a decline, from the highest 368 °C to the lowest 349 °C. Studies on various glasses indicate that a higher ΔT value corresponds to greater resistance to crystallization and increased stability during formation [10]. The phenomenon may be due to the six-coordinated $[MgO_6]$ octahedral structure formed by Mg^{2+} in the glass network, while the twelve-coordinated $[BaO_{12}]$ polyhedral structure formed by Ba^{2+} in the network, leads to the establishment of more complex structural units [20]. Therefore, as MgO gradually replaces BaO in the samples, the internal structural strength of the glass network decreases. The addition of MgO causes irreversible damage to the structure, which increases the amount of non-bridging oxygen and leads to the decrease of the thermal stability of the glass.

Table 2. Thermal characteristics of the five samples.

Glass	Tg (°C)	Tc (°C)	Δ T (°C)
H40	460	828	368
H41	455	813	358
H42	453	807	354
H43	452	809	357
H44	447	796	349

The intricate structures of aluminum borosilicate glasses arise from the presence of two types of glass network formers (SiO_2 and B_2O_3) along with one glass intermediate (Al_2O_3), wherein Al^{3+} may have three coordination environments, namely [AlO_4], [AlO_5] and [AlO_6], and B^{3+} is present in both [BO_3] and [BO_4]. To comprehend the glass structure prepared in this work, both infrared and Raman spectroscopy are employed for analysis [21].

As depicted in Fig. 4(c), the aluminum borosilicate glass has six obvious peaks in the infrared spectrum range of 400–4000 cm^{-1}. The presence of B_2O_3 as the network-forming oxide in all samples result in the presence of three boron-related functional groups in the glass network. The peaks observed at 1340-1472 cm^{-1} are ascribed to the asymmetric stretching vibration of B-O in the [BO_3] unit [22,23]. The absorption peaks at 850-1250 cm^{-1} are associated with the stretching vibration of the Si-O-Si in [SiO_4] tetrahedra and B-O-B linkages in [BO_4] [24,25]. The focus in this region is on the bonding between tetrahedrons. The overlapping of distinct chemical states of Si-O-Si bonds leads to the formation of a broad peak and a significant area in this particular region. The bending vibration of B-O-B in [BO_3] units [26,27] is somewhere between 650 and 730 cm^{-1}. The peaks present at 460-500 cm^{-1} are assigned to the bending vibration of Si-O-Si and Si-O-Al asymmetric bending [28,29]. The peak at 1260 cm^{-1} corresponds to the asymmetric stretching of nonbridging oxygen atoms (NBOs) in trigonal [BO_3] units [16]. Peaks ranging from 3407 cm^{-1} to 3456 cm^{-1} are linked to the symmetric stretching of the O-H bonds [22,30,31]. In summary, the FTIR of the five groups of glasses are roughly similar, indicating that the structure is primarily constituted by [SiO_4] tetrahedron, [BO_4] tetrahedron and [BO_3] triangle. With an increase in the MgO content, the absorption peaks in several regions decrease. Compared with the sample without MgO doping (H40), the infrared absorption of the sample with the largest MgO doping amount (H44) is significantly reduced. This implies that the incorporation of MgO destroys the structural integrity of the SiO_2-B_2O_3 glass network, leading to a decrease in the abundance of [SiO_4], [BO_4] and [BO_3] units, and relaxes the glass structure. The interaction between the structural units is weakened, ultimately causing a reduction in the absorption peak intensity [24].

The intricate polyhedron structure embedded within the glass network and the interconnections among oxygen atoms are unveiled through meticulous analysis using Raman spectroscopy. The Raman spectra of the five samples are plotted in Fig. 4(d), covering the wavenumber range from 100 to 1500 cm^{-1}. The low-frequency region of 200–600 cm^{-1} exhibits numerous spectral bands centered around 460 cm^{-1}. The dominant spectral peak at approximately 890 cm^{-1} in the mid-frequency range of 800–1200 cm^{-1} represents a highly revelatory Raman band, exhibiting a significantly elevated intensity. The study conducted by Mysen et al. [32] demonstrated that this spectral band corresponds to the symmetric stretching vibration of Si-O_{nb}, which arises from a combination of various Q_i species revolving around SiOTs with varied linkages, exhibiting the prominent microscopic structure attributes of borosilicate glass. The observed peak in the spectral range of low frequencies is ascribed to symmetric bending vibration taking place within the bridging oxygen's Si-O-Si chain [33,34]. The feeble spectral peak is detected in the high-frequency band of 1200–1400 cm^{-1} around 1275 cm^{-1}, it is considered to be the B-O asymmetric stretching vibration in [BO_4] [5,11], the vibration peak gradually weakens, indicating that a gradual reduction in the abundance of [BO_4] units present in the glass.

3.2. Thermal properties

The glass powder was blended with a suitable quantity of deionized water to create a paste, which was subsequently pressed into a cylindrical shape by using a mold. The resulting cylinder was then placed on a Kovar base and characterized by HSM. The six temperature characteristic images of the five samples are depicted in Fig. 5, unveiling that with the escalation of furnace temperature, the specimen undergoes contraction and eventually melts at the interface. More precisely, when the amount of MgO substitution gradually rises from 0 wt% to 8 wt%, the sphere points of the glass exhibit a pattern of initial ascent followed by subsequent descent. The halfsphere points, however, display a more complex trend: they first decline from 966 °C to 961 °C, then increase to 970 °C, before decreasing again to 952 °C. Concurrently, the melting point of the glass decreases from the highest value of 1081 °C to the lowest of 968 °C. The above results show that increasing the content of MgO is advantageous for lowering the melting temperature and enhancing the sintering performance of borosilicate.

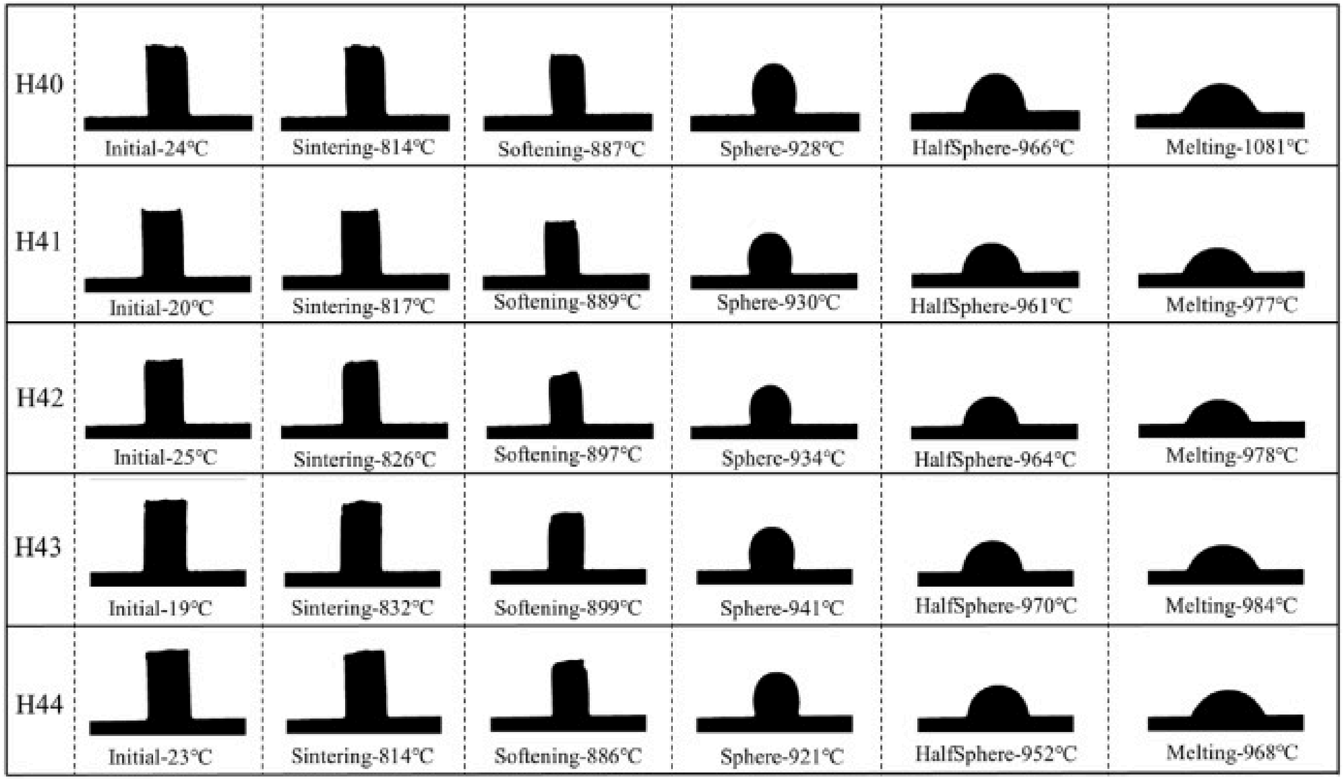

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Fig. 5. HSM image of the glass.

Fig. 6 reveals the sintering shrinkage curves of the glass samples under HSM. The curves depicting the change in volume ratio of the sample throughout the sintering process is portrayed in Fig. 6(a). The volume of the samples slightly increases between 25 °C and 780 °C. However, when the temperature reaches around 780 °C, the sample begins sintering and experiences a reduction in volume. At about 900 °C, volume accomplishes maximum shrinkage, reaching the sintering point of the glass samples. The volume curve of the sample exhibits an upward trajectory as the temperature continues to rise, which is due to the slight expansion of the sample caused by the softening of the outer layer throughout the sintering process. It is worth noting that in low-melting glass powder, the volume may begin to decrease just after reaching Tg [35]. However, in high-melting glass powder, the phenomenon of volume reduction generally does not appear immediately when the temperature reaches Tg [5,11]. In fact, due to the thermal hysteresis effect of glass and the viscosity properties that change with temperature, the process of volume shrinkage is often delayed. Fig. 6(b) is the height ratio curve of the sample during the sintering process. The height ratio reflects the balance between gravity, viscosity and surface tension during the melting process of the glass, which corresponds to the sintering process. The gentle trend of the sample height at about 900°C corresponds to the process of slight expansion of the sample volume, and then the rapid decrease of the height is attributed to the low viscosity of the sample at this temperature, which leads to the destruction of the balance between viscosity, gravity and surface tension, and accelerates the transformation of the sample into molten fluid. The results indicate that all the glass samples can complete the sintering process before 900°C, and when the substitution amount of MgO is 8 wt%, the sample obtains the best sintering performance.

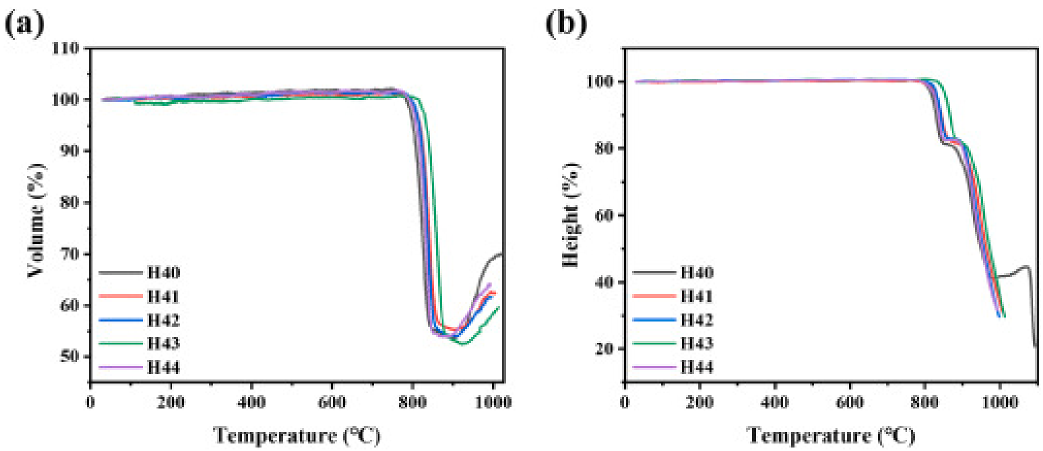

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Fig. 6. (a) Volume shrinkage curves; (b) Hight shrinkage curves.

3.3. Wettability

The contact angle is defined as the angle at which the three-phase boundary, starting from within the liquid and extending to the interface between gas and liquid, denoted by θ. Typically, the θ is utilized as a macroscopic indicator to determine the wettability of a surface [36]. During the wetting process, with increasing temperature, the glass gradually reaches its softening temperature, leading to a reduction in the viscosity of the molten glass. As the molten glass wets and spreads on the metal surface, the interfacial tension between the liquid and gas phases decreases. The solid-liquid droplet experiment is commonly employed to investigate the solid-liquid interface. The morphology of the wetting flow interface between glass and metal surface is determined by the interaction of gas, liquid, and solid three-phase interface energy.

The change of contact angle with temperature during melting of glass samples was recorded by HSM, as depicted in Fig. 7. When the sample reaches its sintering temperature (about 800°C), the metal fails to wet the glass and thus cannot effectively spread on its surface. The sample's volume remains constant, so the contact angle tends to be stable. As the temperature continues to increase, the curve shows a downward trend, indicating that the glass has reached the sintering point at that time, and the volume begins to shrink, leading to a reduction in the θ. After that, the curve shows an increasing trend at about 900 °C, corresponding to the expansion phenomenon in the glass sintering process, causing an increase in the θ. Upon attaining its melting point, the liquid glass is completely spread on the metal surface, and the θ gradually decreases within the range of 0°<θ<90° until it is finally in equilibrium. The secondary increase of the H40 sample also corresponds to its height ratio curve.

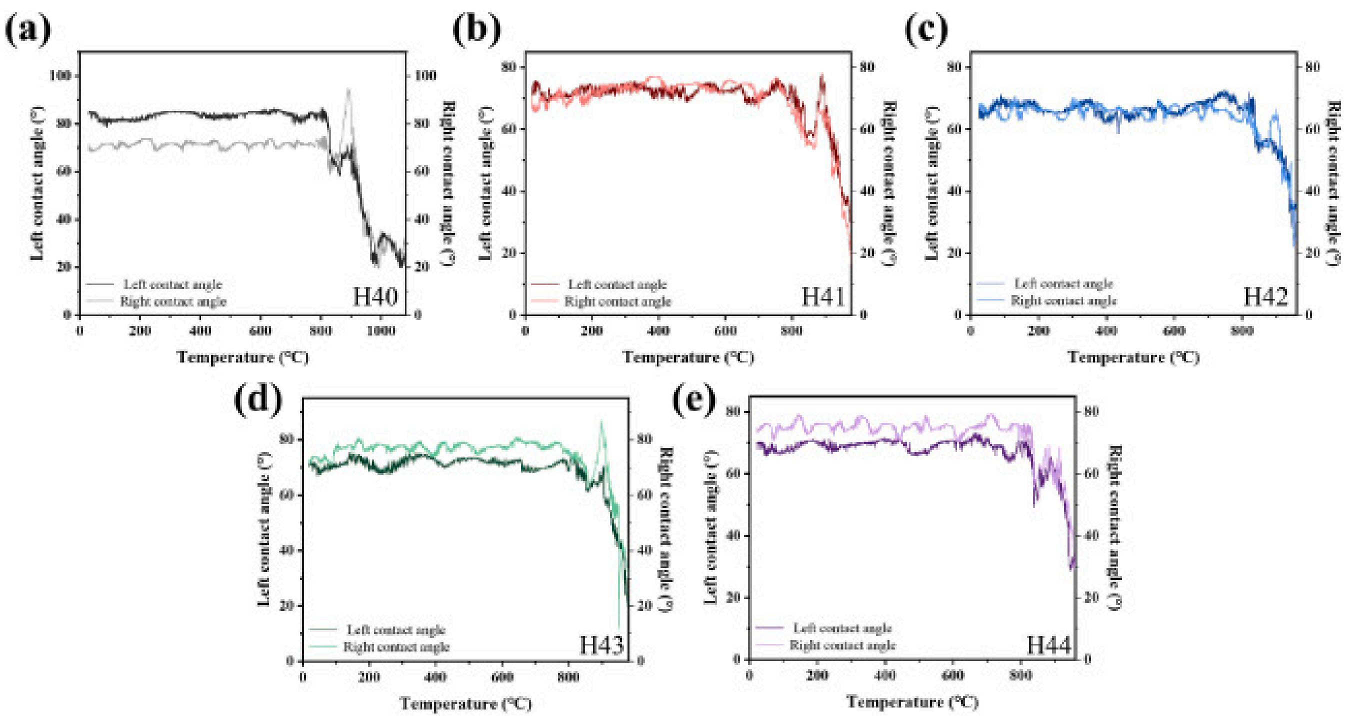

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Fig. 7. (a–e) Left and right contact angle curves of the five glass samples with Kovar as a function of temperature.

Contact angle meter was used to measure the final melted samples. The contact angles of the samples show a progressive growth with the increase in MgO content, as depicted in Fig. 8(b–f), which can be attributed to the augmented viscosity of the glass. Consequently, the wettability of the Kovar alloy substrate gradually decreases.

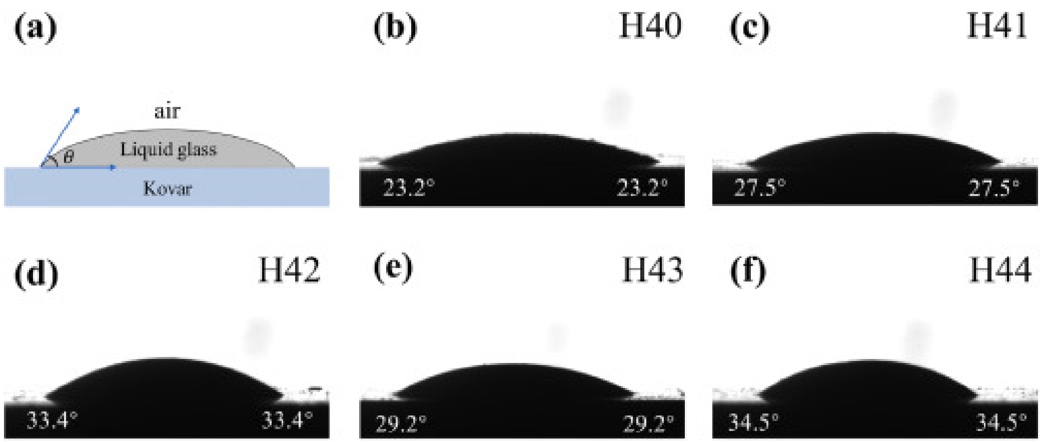

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Fig. 8. (a) Contact angle model; (b–f) Contact angle of the five samples with Kovar.

3.4. Permeability performances

Fig. 9 shows the changes of light absorption, transmission and reflection spectra of the glass sheets across a wavelength range of 230–2600nm. Fig. 9(b) illustrates that ultraviolet light in the 230–350nm band is almost completely reflected. Between 350 and 500nm, the transmittance of the glass sheets gradually increases while the reflectivity decreases. The glass sheets exhibit a transmittance of over 70% across most wavelengths and exceed 80% in the visible light spectrum, indicating good glass formation. In the near-infrared band of 700–1500nm, the transmittance decreases due to reflection and weak absorption, which might be characteristic of this glass system. In this range, the glass reflects more light. The transmittance of the glass reveals an increasing trend across the entire spectrum with the augmentation of MgO content, indicating that an optimal concentration of MgO can enhance the clarity of the glass, help to reduce bubbles and other impurities in the glass, and improve its transparency and optical characteristic. The actual photos of the prepared glass sheets were shown in Fig. 9.

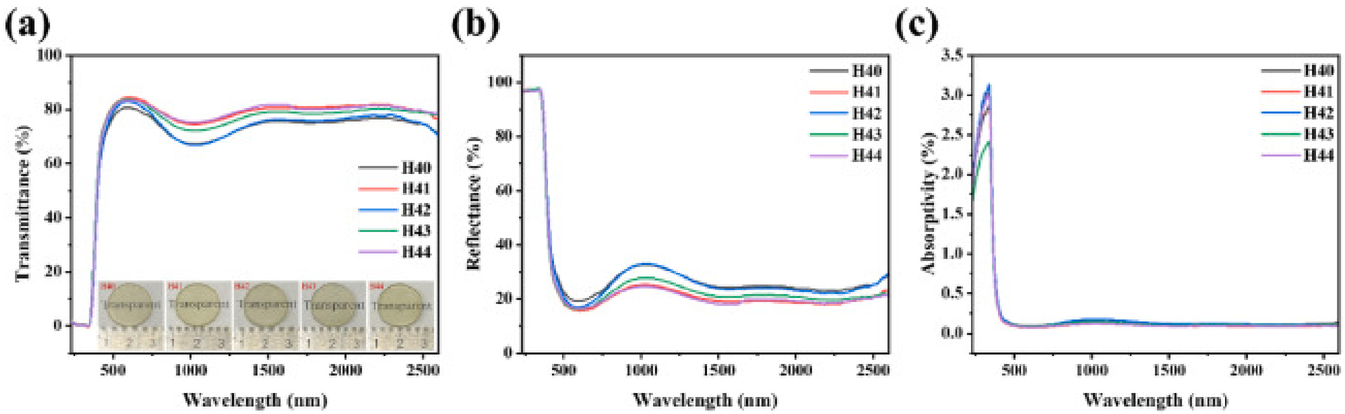

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Fig. 9. (a) Transmittance curves; (b) Reflectance curves; (c) Absorptivity curves.

3.5. Dielectric properties

The dielectric constants of the glasses are decided by the contribution of electron orientation, ion orientation and dipole orientation to the polarizability. Fig. 10(a) displays the variation in dielectric constants for different MgO/BaO compositions within a frequency range of 10^0–10^6Hz. Demonstrating a gradual decrease in ε_r as the frequency increases, this phenomenon can be assigned to the enhanced polarizability resulting from increased ionic orientation mobility within glass systems. The period of ionic alternating motion at low frequencies becomes significantly larger and similar to the occurrence rate of the oscillating electrical field, resulting in a consequent increase in dielectric constant. However, upon frequency escalation, the ability of ionic alternating motion to move sufficiently and rapidly diminishes. The ionic oscillation, as a consequence, will lag behind the changes in the electric field, ultimately resulting in a reduction of the dielectric constant [37].

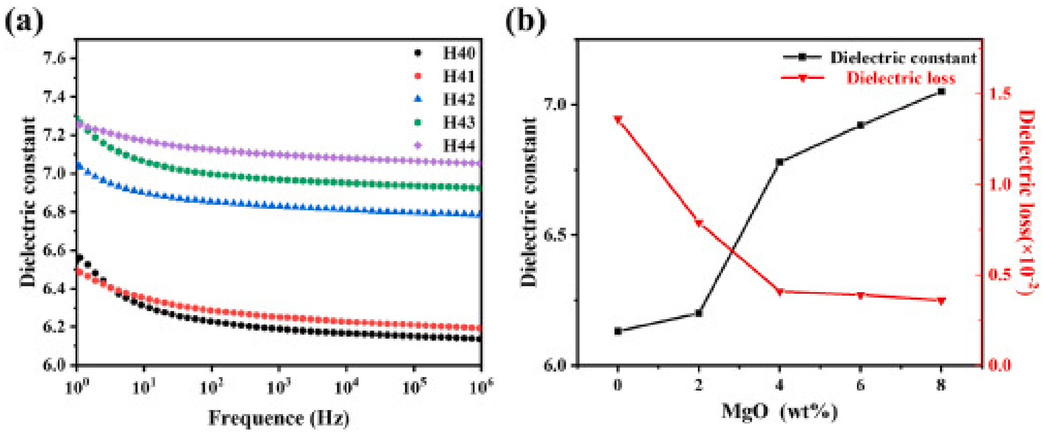

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Fig. 10. (a) The relation between dielectric constant and frequency; (b) The dielectric properties of all samples at 1 MHz.

The dielectric properties of samples at 1 MHz are depicted in Fig. 10(b). The dielectric constants are found to exhibit an upward trend. This phenomenon can be ascribed to the incorporation of MgO, resulting in an increase in non-bridging oxygen atoms and causing a disruption to the glass network structure. First, the dielectric constant experiences notable alterations due to the mobility of modifying ions within the spaces between them, as evidenced by numerous studies. The disrupted structure can enhance the mobility of Ba^{2+} and Mg^{2+} ions, thus augmenting ionic polarization [38]. Second, under the action of an electric field, dipoles in glass will reorient to be consistent with the electric field direction. When the network structure of glass is loose, the activity of dipoles increases, and they are more likely to reorient under the action of an electric field, thus leading to an increase in the dielectric constant [39]. Third, the distortion of the glass network leads to a rise in bonding imperfections, thereby enhancing polarization caused by the presence of space charge, and resulting in an elevated dielectric constant [40]. Therefore, as the substitution of BaO with MgO, the internal structure becomes loose. This results in a decrease in the glass's stability and causes an elevation in the dielectric constant. The dielectric loss of glass is primarily associated with the displacement polarization of electrons, ion displacement polarization, microstructure polarization and dipole polarization. In addition, it is evident from the Raman spectra that the 890 cm^{-1} Raman band corresponds to the vibration of Si-O_{nb} in a symmetrical stretching manner within the glass network, which arises from a combination of various Qi species with different connection modes centered on SiOTs. The vibration induces changes in the degree of polarization and leads to a Raman shift in the energy exchange of incident photons. Due to their location within the tetrahedron, alkaline earth cations decrease the quantity of bridging oxygen atoms, the Si-O-Si asymmetric tensile vibrations experience a reduction in strength. The correlation between the dielectric loss and network vibration is highly close. A diminished dielectric loss in the glass material is induced by the movement of the vibration peak and the attenuation of the Si-O-Si asymmetric stretching mode.

3.6. CTE, densities and shear resistances

Fig. 11 depicts the flow chart of shear force test, the CTE curves, density histogram and shear force curves of the five groups of glass samples. The parameters of CTE and swelling softening temperature point (Ts) are extracted by using the linear expansion curve dL/L_0(%) with the change of temperature [41,42]. The linear CTE values are consistently decided within the range of 25–500 °C, with Ts representing the peak temperature in the expansion region. Table 3 summarizes the CTE, Ts and shear strength of all samples. It is observed that the Ts of the glass exhibits minimal variation, while the CTE gradually increases from the 7.40×10^{-6}/°C to 8.00×10^{-6}/°C. The density of the glass system exerts a primary influence, as evidenced by the gradual decrease in density from 2.832 g cm^{-3} to 2.679 g cm^{-3} with an increasing MgO content, as illustrated in Fig. 11(c). However, the density is mostly determined by the network structure [43]. The denser the structure, the smaller the CTE. After examining the thermal stability and dielectric characteristics, the observation reveals that as the MgO substitution increases, there is a gradual relaxation in the network structure, leading to a corresponding rise in CTE.

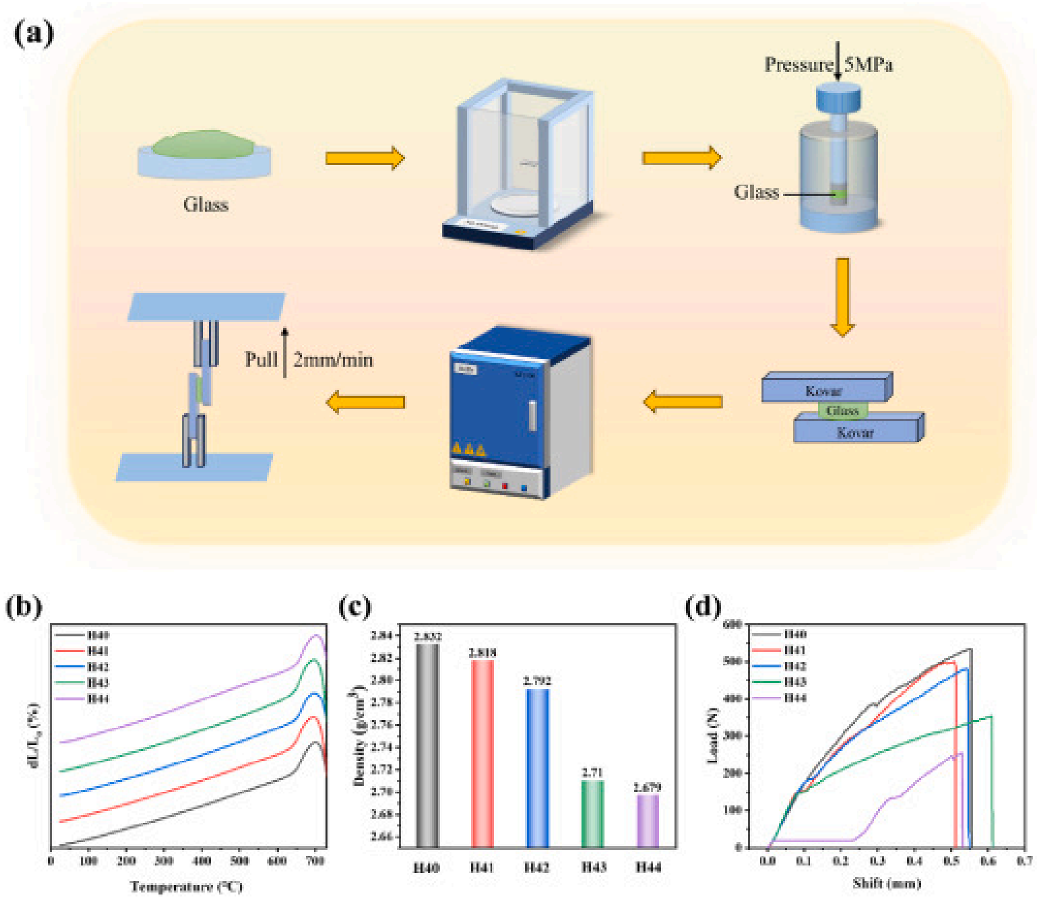

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Fig. 11. (a) Flow chart of shear force test; (b) CTE curves; (c) Density histogram; (d) Shear force curves.

Table 3. CTE, Ts and Shear forces of the five glass samples.

Sample	**CTE$_{(500°C)}$($\times 10^{-6}$/°C)**	**Ts (°C)**	**Shear force (N)**
H40	7.40	701	534
H41	7.60	693	500

Sample	$CTE_{(500°C)}$ ($\times 10^{-6}$/°C)	Ts (°C)	Shear force (N)
H42	7.64	698	481
H43	7.74	697	354
H44	8.00	703	255

An equal amount of glass powder was pressed into an 8 mm mold. The mechanical properties of Kovar alloy brazed with glass samples were measured under the welding environment of 1000 °C for 5 min. Fig. 11(d) depicts the shear force curves of the specimen throughout the experiment. The existence of the platform stage indicates that the glass has a certain toughness, and the other is brittle fracture. The analysis curve shows that the shear force of all samples reaches more than 250 N, with good adhesion. As MgO substitution increases, the shear force of the sample gradually decreases, from the maximum 534 N to the minimum 255 N. The corresponding shear strength can be reduced from 10.63 MPa to 5.08 MPa. One explanation could be the gradual relaxation of the glass structure, leading to reduced stability and diminished shear force. Secondly, the investigation shows that the CTE of the Kovar alloy is about 5.5-6.5$\times 10^{-6}$/°C, and as the CTE of the sample gradually increases, its matching degree with the Kovar alloy becomes weaker. As a result, the shear force of the seal is gradually reduced. The CTE of the glass used as sealant in Kovar alloy is generally between 5 and 8$\times 10^{-6}$/°C, and the values of the glass prepared in this work are all in this range, which indicates that they are suitable for the encapsulation of Kovar alloy devices.

4. Conclusions

This study investigated the impact of gradually replacing BaO with MgO on the structure, thermal, optical, and dielectric properties of borosilicate glass. With the increase in MgO concentration, both the Tg and Tc decreased, indicating a loosening of the glass structure and reduced resistance to crystallization. FTIR and Raman analyses confirmed that the glass network, which is primarily composed of [SiO_4], [BO_4], and [BO_3] units, became disrupted with higher MgO content, leading to the formation of non-bridging oxygens and a consequent decline in structural stability. The dielectric constant increased from 6.13 to 7.05, attributed to enhanced cation migration and ion polarization within the loosened structure. Additionally, the CTE rose from 7.40$\times 10^{-6}$/°C to 8.00$\times 10^{-6}$/°C, while shear strength decreased from 534 N to 255 N, correlating with the reduced glass density. Compared to other samples, H41 exhibited a lower melting point and dielectric constant, as well as superior shear performance.

Consequently, we compared H41 with data from previous relevant studies (Table 4), and its performance metrics remain very competitive. This work provides valuable insights into the influence of alkaline earth metal oxides (MgO and BaO) on the properties of borosilicate glass, offering new approaches for material design and application in heat-resistant glass and electronic packaging.

Table 4. Parameter comparison between H41 and other glass materials.

Sample	CTE ($\times10^{-6}$/°C)	Shear strength (MPa)	Tg (°C)	Reference
SiO_2-SrO-RO	8.5–11	–	608–637	[6]
SiO_2-Al_2O_3-RO	6.64–9.28	–	432–675	[9]
B_2O_3-SiO_2-RO	7.89–8.27	–	583–600	[11]
CaO- B_2O_3-RO	11.5	1.16–4.47	–	[13]
H41	7.60	9.95	455	This work

CRediT authorship contribution statement

Lianghong He: Writing – original draft. **Yunhao Fu:** Investigation. **Yuxin Tian:** Investigation. **Jintao Bai:** Funding acquisition, Formal analysis. **Shenghua Ma:** Writing – review & editing, Conceptualization. **Gang Wang:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations, there is no professional or other personal conflicts of interest of any nature or kind in any product.

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Appendix A. Supplementary data

The following is the Supplementary data to this article:

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Multimedia component 1.

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